Windows Scheduled Task Persistence Investigation

Controlled Windows 11 SOC Lab — Scheduled Task Creation, Launch Telemetry & Execution Validation

Executive Summary

This investigation examined a controlled Windows scheduled-task persistence simulation. A harmless scheduled task named SOC-Lab-Discovery-Test was created to execute C:\Windows\System32\whoami.exe under the SOCAnalyst2 interactive user context. Windows Security Event ID 4698 and Task Scheduler Operational Event 106 confirmed task creation/registration at 01/09/2026 20:43:26. A manual run at 20:56:20 generated Task Scheduler Operational Events 110 and 325, confirming that the task instance was launched and queued. The investigation deliberately distinguishes confirmed evidence from inference. Although Get-ScheduledTaskInfo reported LastTaskResult = 0 and recorded LastRunTime as 20:56:20, neither Task Scheduler Operational telemetry nor Sysmon independently confirmed creation of a whoami.exe process. The final assessment therefore treats the actual process execution as unconfirmed.

Lab Environment

Host/VM: Windows 11 virtual lab using Oracle VirtualBox. Sysmon: Installed and running as service Sysmon. Primary lab account: SOCAnalyst2. Scheduled task: SOC-Lab-Discovery-Test. Configured action: C:\Windows\System32\whoami.exe. Task principal: SOCAnalyst2, Interactive logon, Limited run level. Task settings: Enabled; AllowDemandStart=True; MultipleInstances=IgnoreNew; RunOnlyIfIdle=False; RunOnlyIfNetworkAvailable=False.

Investigation Objective

Determine whether a scheduled task could be identified as a persistence mechanism through Windows Security and Task Scheduler telemetry, then attempt to correlate its launch with endpoint process telemetry from Sysmon. The objective was to build an evidence-based timeline and explicitly document any telemetry gaps.

Telemetry Preparation

Task Scheduler Operational logging was enabled with: wevtutil sl Microsoft-Windows-TaskScheduler/Operational /e:true Auditing for Other Object Access Events was enabled with: auditpol /set /subcategory:"Other Object Access Events" /success:enable /failure:enable Sysmon was verified running under the service name Sysmon, with telemetry available in Microsoft-Windows-Sysmon/Operational.

Evidence Timeline

20:43:26 — Security Event ID 4698: A scheduled task was created. 20:43:26 — Task Scheduler Operational Event 106: SOCAnalyst2 registered \SOC-Lab-Discovery-Test. 20:56:20 — Task Scheduler Operational Event 110: Task Scheduler launched an instance of \SOC-Lab-Discovery-Test for SOCAnalyst2. 20:56:20 — Task Scheduler Operational Event 325: The task instance was queued. 20:56:20 — Get-ScheduledTaskInfo: LastRunTime recorded as 20:56:20 and LastTaskResult reported 0. After 20:56:20 — No Event 200/201 action-launch/completion telemetry was found for this task, and no Sysmon Event ID 1 for whoami.exe was observed.

Key Findings

Finding 1 — Scheduled task creation was confirmed. Security Event ID 4698 identified the creation event, while Task Scheduler Event 106 recorded registration by SOCAnalyst2. Finding 2 — The task was configured with a benign discovery command. Its action was C:\Windows\System32\whoami.exe, with no arguments, under an interactive limited user context. Finding 3 — A subsequent task launch/queue operation was confirmed at 20:56:20 through Task Scheduler Events 110 and 325. Finding 4 — Actual process execution was not independently confirmed. No matching Task Scheduler Event 200/201 or Sysmon Event ID 1 for whoami.exe was identified. Finding 5 — The scheduler's LastTaskResult value of 0 should not be treated as sole proof of process execution. The investigation records this as a telemetry discrepancy rather than resolving it by assumption.

Detection & SOC Relevance

Scheduled tasks can provide persistence because they can cause programs to execute automatically according to triggers or other conditions. From a SOC perspective, useful detection opportunities include newly created tasks, unusual task names or paths, suspicious executables or scripts in task actions, unexpected principals or privilege levels, and process creation correlated to task-launch telemetry. In this controlled lab, the strongest observable indicators were Security 4698 and Task Scheduler Operational 106, followed by the later 110/325 launch/queue events. Process telemetry should be used for correlation when available, but absence of a Sysmon process event should be recorded as a visibility limitation rather than interpreted as proof that no execution occurred.

MITRE ATT&CK; Mapping

T1053.005 — Scheduled Task/Job: Scheduled Task. This mapping is directly relevant because the lab activity created and launched a Windows scheduled task. The lab also used whoami.exe as a benign system/user discovery action, but because actual whoami.exe process execution was not independently confirmed, the investigation does not treat that technique as confirmed execution evidence.

Limitations & Telemetry Gaps

The available evidence confirms task creation, registration, and a later launch/queue event. It does not provide independent proof that whoami.exe created a process. The task remained reported as Queued during investigation, and no subsequent task-specific action completion event was identified. This is an important analytical limitation. A SOC analyst should avoid converting scheduler status fields into unsupported claims about endpoint process execution.

Conclusion

The controlled lab successfully demonstrated how scheduled-task creation and launch activity can be investigated using Windows Security and Task Scheduler logs. The investigation confirmed the persistence mechanism and reconstructed a defensible timeline. It also demonstrated the importance of validating execution across independent telemetry sources. Final assessment: BENIGN / CONTROLLED LAB ACTIVITY. No malicious behavior was established. The primary investigative lesson is evidence correlation: task creation and scheduler activity were confirmed, while the configured process execution remained unconfirmed by independent endpoint telemetry.

Evidence Table

Time	Source	Evidence
20:43:26	Security 4698	Scheduled task created
20:43:26	Task Scheduler 106	Task registered by SOCAnalyst2
20:56:20	Task Scheduler 110	Task instance launched
20:56:20	Task Scheduler 325	Task instance queued
20:56:20	ScheduledTaskInfo	LastRunTime; LastTaskResult = 0
Post-run	Task Scheduler / Sysmon	No independent whoami.exe execution observed